

10/04/2025

CODE-A



Aakash

Medical | IIT-JEE | Foundations

Corporate Office: AESL, 3rd Floor, Incuspaze Campus-2, Plot-13,
Sector-18, Udyog Vihar, Gurugram, Haryana-122015



AIM - 720

(Advanced INTENSIVE Mastery for 720)

CC-108

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MM : 720

CST-2

Time : 180 Mins.

Complete Syllabus of NEET

Instructions:

- Duration of Test is 3 hrs.
- The Test consists of 180 questions. The maximum marks are 720.
- There are three parts in the question paper consisting of Chemistry, Physics, Biology having 45 questions each in Physics and Chemistry and 90 questions in Biology).
- Each question carries +4 marks. For every wrong response, -1 mark shall be deducted from the total score. Unanswered/unattempted questions will be given no marks.
- Use blue/black ballpoint pen only to darken the appropriate circle.
- Mark should be dark and completely fill the circle.
- Dark only one circle for each entry.
- Dark the circle in the space provided only.
- Rough work must not be done on the Answer sheet and do not use white fluid or any other rubbing material on the Answer sheet.

6 Aakashians scored a perfect 720/720 from Classroom Programs



720/720 | **Mridul M Anand**
3 Year Classroom



720/720 | **Ayush Naugraiya**
4 Year Classroom



720/720 | **Arghyadeep Dutta**
2 Year Classroom



720/720 | **Aryan Yadav**
1 Year Classroom



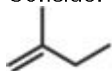
720/720 | **Palansha Agarwal**
2 Year Classroom



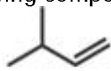
720/720 | **Iram Quazi**
1 Year Classroom

CHEMISTRY

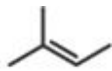
1. Consider the following compounds



(i)



(ii)



(iii)

Correct order of energy released on catalytic hydrogenation of the given compound is

- (1) (ii) > (i) > (iii)
 - (2) (iii) > (ii) > (i)
 - (3) (iii) > (i) > (ii)
 - (4) (ii) > (iii) > (i)
2. Ethene on reaction with Baeyer's reagent majorly gives
- (1) Ethanal
 - (2) Ethylene glycol
 - (3) Ethanol
 - (4) Glyoxal

3. Match List-I (IUPAC nomenclature) with List-II (IUPAC Official Name),

	List-I		List-II
a.	Unniltrium	(i)	Seaborgium
b.	Unnilhexium	(ii)	Nobelium
c.	Unnilquadium	(iii)	Lawrencium
d.	Unnilbium	(iv)	Rutherfordium

Choose the correct option.

- (1) a(ii), b(iv), c(i), d(iii)
 - (2) a(i), b(iii), c(iv), d(ii)
 - (3) a(iii), b(i), c(iv), d(ii)
 - (4) a(iii), b(i), c(ii), d(iv)
4. Identify the group and period respectively, for an element with atomic number 113.
- (1) 13 and 6
 - (2) 13 and 7
 - (3) 12 and 7
 - (4) 14 and 6
5. Standard reduction potential of three metal electrodes ($E^\circ_{M^{n+}/M}$) A, B and C respectively are 0.22 V, -0.14 V and 0.25 V. Identify the correct order of reducing power for these metals.
- (1) A > B > C
 - (2) B > A > C
 - (3) C > A > B
 - (4) A > C > B

6. Given below are two statements one is labelled as **Assertion (A)** and other is labelled as **Reason (R)**

Assertion (A) : CF_3COOH is more acidic than CCl_3COOH

Reason (R) : 'F' has higher electron withdrawing effect than 'Cl'

In the light of above statements, choose the most appropriate answer from the option given below.

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A)
 - (2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
 - (3) (A) is true but (R) is false
 - (4) (A) is false but (R) is true
7. If $t_{1/2}$ of a zero order reaction is 10 min then $t_{75\%}$ will be
- (1) 20 min
 - (2) 15 min
 - (3) 12 min
 - (4) 18 min
8. Under isothermal condition, a gas at 300 K expands from 0.1 L to 0.6 L against a constant external pressure of 1.5 bar. The work done by the gas is
- (1) -75 J
 - (2) +75 J
 - (3) -2.25 J
 - (4) +2.25 J
9. Consider the following statements:
- (i) Mass of an atom of hydrogen is 1.6736×10^{-24} g
 - (ii) N_2 and O_2 contain same number of nucleons.
 - (iii) Compounds are formed when two or more atoms of different elements combine together in a definite ratio.
- Choose the correct statement(s).
- (1) (i) only
 - (2) (i) and (ii) only
 - (3) (ii) and (iii) only
 - (4) (i) and (iii) only
10. Which of the following statement about H_2SO_4 is incorrect?
- (1) It can act as an oxidising agent with copper.
 - (2) It can act as a dehydrating agent with sugar.
 - (3) V_2O_5 acts as a catalyst in contact process for manufacturing H_2SO_4
 - (4) It can act as a reducing agent in its reaction with sulphur

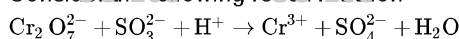
11. The gas evolved on reaction of gold with aqua regia is

- (1) N_2O
- (2) NO
- (3) NO_2
- (4) N_2

12. The correct IUPAC name for Phenyl isopentyl ether is

- (1) 3-Methylbutoxybenzene
- (2) 2-Methylbutoxybenzene
- (3) Pentoxybenzene
- (4) Phenoxy-3-methylbutane

13. Consider the following redox reaction



Match the reactants/products given in List-I with the stoichiometric coefficients given in List-II.

List-I		List-II
a. $Cr_2O_7^{2-}$	(i)	8
b. SO_3^{2-}	(ii)	4
c. H^+	(iii)	1
d. H_2O	(iv)	3

Choose the correct option containing most appropriate match

- (1) a(iii), b(iv), c(ii), d(i)
- (2) a(iii), b(iv), c(i), d(ii)
- (3) a(iv), b(iii), c(ii), d(i)
- (4) a(iv), b(iii), c(i), d(ii)

14. Consider the following statements.

- (i) When urea is heated with concentrated sulphuric acid, then nitrogen present in it, gets converted into ammonium sulphate.
- (ii) Kjeldahl's method is not applicable to compounds containing nitrogen in nitro and azo group.
- (iii) Nitrogen present in pyridine can be estimated by Kjeldahl's method.

The incorrect statement(s) is/are

- (1) (i) & (iii) only
- (2) (i) only
- (3) (iii) only
- (4) (i), (ii) & (iii)

15. Which of the given compounds show metamerism?

- (1) C_5H_{10}
- (2) $C_4H_{10}O$
- (3) C_3H_8O
- (4) C_2H_6O

16. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : SiF_6^{2-} exists but $SiCl_6^{2-}$ does not.

Reason (R) : Hybridisation of Si in SiF_6^{2-} is sp^3d^2

In the light of above statements, choose the most appropriate answer from the option given below.

- (1) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
- (2) Both Assertion (A) and Reason (R) are true but Reason (R) is not the correct explanation of Assertion (A).
- (3) Assertion (A) is true and Reason (R) is false.
- (4) Assertion (A) is false and Reason (R) is true.

17. Match species given in List-I with Hybridisation of central atom given in List-II.

List-I		List-II
a. XeO_3	(i)	sp^2
b. $[Ni(CN)_4]^{2-}$	(ii)	sp^3d^2
c. SF_6	(iii)	dsp^2
d. CO_3^{2-}	(iv)	sp^3

The correct match is

- (1) a(i), b(ii), c(iii), d(iv)
- (2) a(iii), b(iv), c(i), d(ii)
- (3) a(iv), b(iii), c(ii), d(i)
- (4) a(ii), b(i), c(iii), d(iv)

18. The correct order of bond angle for the following species is

- (1) $BeCl_2 > NH_3 > CH_4 > H_2O > BF_3$
- (2) $BeCl_2 > BF_3 > CH_4 > NH_3 > H_2O$
- (3) $H_2O > NH_3 > CH_4 > BF_3 > BeCl_2$
- (4) $CH_4 > NH_3 > BF_3 > BeCl_2 > H_2O$

19. Consider the following statements.

Statement I : n-Hexane and n-heptane solution shows positive deviation from Raoult's law.

Statement II : A mixture of chloroform and acetone shows negative deviation from Raoult's law.

In the light of above statements, choose the most appropriate option.

- (1) Both statement I and statement II are correct.
- (2) Both statement I and statement II are incorrect.
- (3) Statement I is incorrect but statement II is correct.
- (4) Statement I is correct but statement II is incorrect.

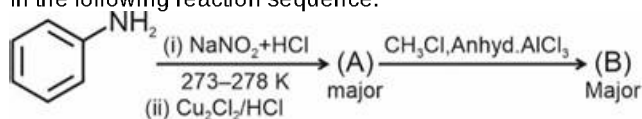
20. Which of the following solution will have minimum freezing point?

- (1) 0.1 m NaCl
- (2) 0.1 m CaCl_2
- (3) 0.1 m Urea
- (4) 0.1 m Glucose

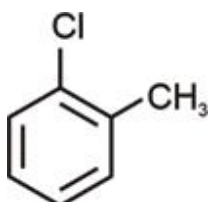
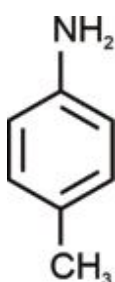
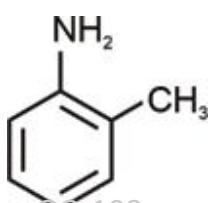
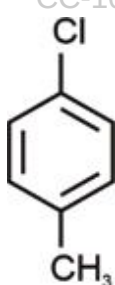
21. Which among the following is a homoleptic complex?

- (1) Pentaamminecarbonatocobalt (III) chloride
- (2) Tetraammineaquachloridocobalt (III) chloride
- (3) Potassium trioxalatochromate (III)
- (4) Dichloridobis (ethane-1, 2-diamine) cobalt (III) chloride

22. In the following reaction sequence.



The correct structure of major product (B) is

- (1) 
- (2) 
- (3) 
- (4) 

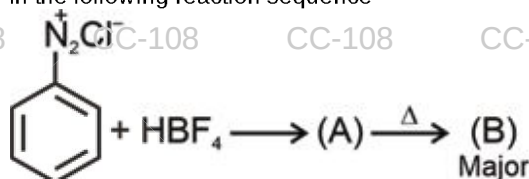
23. Consider the following statements

- (a) Boiling point of 2-Bromo-2-methylpropane is higher than 2-Bromobutane
- (b) Boiling point of 2-bromobutane is higher than 2-Iodobutane
- (c) Boiling point of o-dichlorobenzene is higher than p-dichlorobenzene

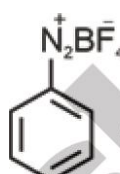
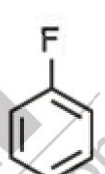
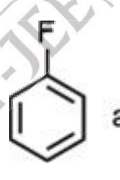
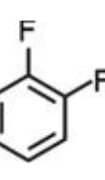
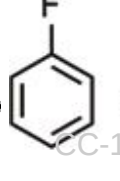
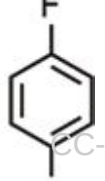
The **incorrect** statement(s) is/are

- (1) (a) and (c) only
- (2) (a) and (b) only
- (3) (b) only
- (4) (a), (b) and (c)

24. In the following reaction sequence



(A) and (B) respectively are

- (1)  and 
- (2)  and 
- (3)  and 
- (4)  and 

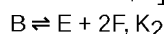
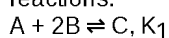
25. Match compounds in List-I with their colour in List-II

List-I	List-II
a. $\text{Fe}_4[\text{Fe}(\text{CN})_6]_3 \cdot x\text{H}_2\text{O}$	(i) Yellow
b. PbS	(ii) Prussian blue
c. $(\text{NH}_4)_3\text{PO}_4 \cdot 12\text{MoO}_3$	(iii) Blood red
d. $[\text{Fe}(\text{SCN})]^{2+}$	(iv) Black

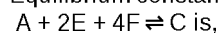
The correct match is

- (1) a(iv), b(ii), c(i), d(iii)
 (2) a(ii), b(iv), c(i), d(iii)
 (3) a(iv), b(ii), c(iii), d(i)
 (4) a(ii), b(iv), c(iii), d(i)

26. Consider the equilibrium constants for the following reactions.



Equilibrium constant (K) for the reaction

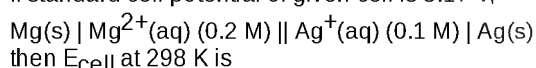


- (1) $\frac{K_2^2}{\sqrt{K_1}}$
 (2) $K_2 K_1^2$
 (3) $K_1 K_2^2$
 (4) $\frac{K_1}{K_2^2}$

27. Calculate the pH of the solution containing a mixture of 50 mL, 0.04 M acetic acid and 50 mL, 0.02 M sodium salt of acetic acid (pK_a of acetic acid = 4.76, $\log 2 = 0.3010$)

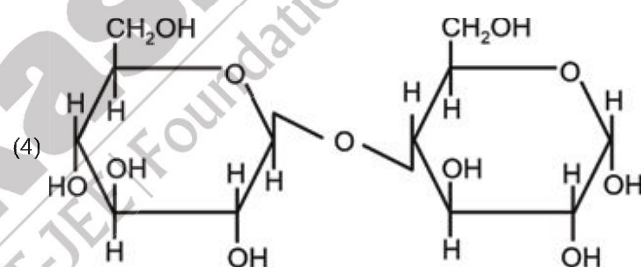
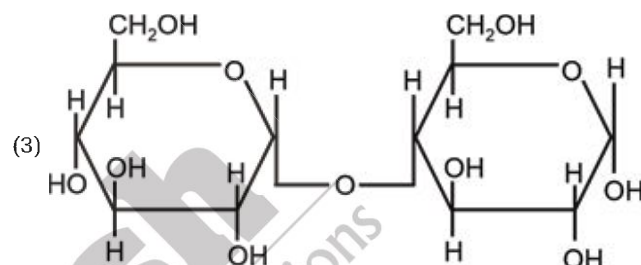
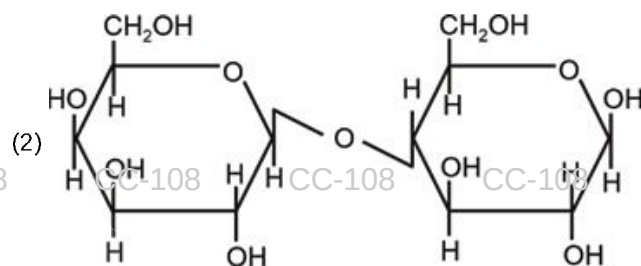
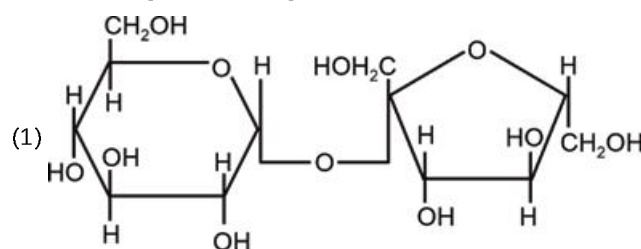
- (1) 9.54
 (2) 8.93
 (3) 4.46
 (4) 5.06

28. If standard cell potential of given cell is 3.17 V,



- (1) 5.63 V
 (2) 2.97 V
 (3) 4.92 V
 (4) 3.13 V

29. Which among the following is a correct structure of lactose?



30. Given below are two statements

Statement I: Naphthalene and Anthracene are aromatic compounds.

Statement II: Naphthalene contains 14 π electrons while Anthracene contains 10 π electrons.

In the light of above statements, choose the most appropriate answer from the option given below.

- (1) Statement I is correct but statement II is incorrect
 (2) Both statement I and statement II are correct
 (3) Both statement I and statement II are incorrect
 (4) Statement I is incorrect but statement II is correct

31. Match List-I with List-II

	List-I Subshells		List-II Radial Nodes
a.	4p	(i)	1
b.	5s	(ii)	2
c.	4d	(iii)	Zero
d.	4f	(iv)	4

The correct match is

- (1) a(ii), b(iv), c(iii), d(i)
 (2) a(iii), b(iv), c(i), d(ii)
 (3) a(iv), b(ii), c(i), d(iii)
 (4) a(ii), b(iv), c(i), d(iii)

 32. For the process, $\text{H}_2\text{O(l)} (1 \text{ bar}, 373 \text{ K}) \rightarrow \text{H}_2\text{O(g)} (1 \text{ bar}, 373 \text{ K})$, the correct set of thermodynamic parameters will be:

- (1) $\Delta G = 0$, $\Delta S_{\text{system}} = 0$
 (2) $\Delta G = 0$, $\Delta S_{\text{system}} > 0$
 (3) $\Delta G < 0$, $\Delta S_{\text{system}} = 0$
 (4) $\Delta G < 0$, $\Delta S_{\text{system}} > 0$

33. Given below are two statements

Statement I: The equilibrium constant for an exothermic reaction decreases as the temperature increases.

Statement II: The equilibrium constant for an endothermic reaction increases as the temperature increases.

In the light of the above statements, choose the most appropriate answer from the options given below.

- (1) Both statement I and statement II are correct
 (2) Both statement I and statement II are incorrect
 (3) Statement I is correct but statement II is incorrect
 (4) Statement I is incorrect but statement II is correct

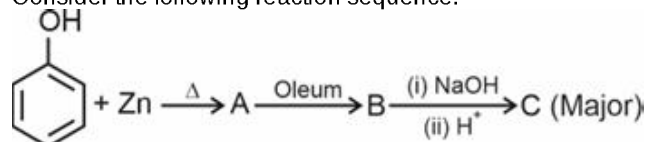
34. Match List-I with List-II

	List-I (Common method of preparation)		List-II (Oxides of nitrogen)
a.	$\text{NH}_4\text{NO}_3 \xrightarrow{\text{Heat}}$	(i)	N_2O_3
b.	$\text{Pb}(\text{NO}_3)_2 \xrightarrow{673 \text{ K}}$	(ii)	N_2O
c.	$\text{NaNO}_2 + \text{FeSO}_4 + \text{H}_2\text{SO}_4 \rightarrow$	(iii)	NO_2
d.	$\text{NO} + \text{N}_2\text{O}_4 \xrightarrow{250 \text{ K}}$	(iv)	NO

Choose the correct answer from the options given below

- (1) a(ii), b(iii), c(i), d(iv)
 (2) a(ii), b(iii), c(iv), d(i)
 (3) a(iii), b(ii), c(iv), d(i)
 (4) a(i), b(iii), c(iv), d(ii)

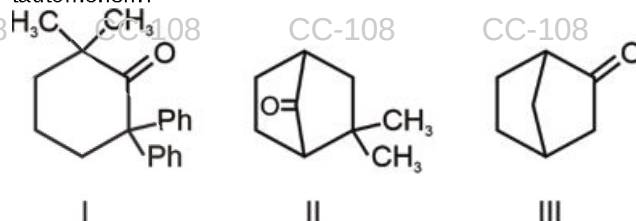
35. Consider the following reaction sequence.



The major product C is

- (1) Benzene sulphonic acid
 (2) Anisole
 (3) Benzaldehyde
 (4) Phenol

36. Which among the given molecules can exhibit tautomerism?



Choose the correct option

- (1) III only
 (2) I & III only
 (3) II & III only
 (4) I, II & III

37. Consider the following statements

 (a) O_2 molecule is paramagnetic in nature.

 (b) N_2^+ has bond order 2.5.

 (c) Li_2^{2+} does not exist

The correct statements are

- (1) (a) and (b) only
 (2) (b) and (c) only
 (3) (a) and (c) only
 (4) (a), (b) and (c)

38. Given below are two statements

Statement I: $[\text{MnBr}_4]^{2-}$ has sp^3 hybridisation and tetrahedral shape

Statement II: The spin only magnetic moment of $[\text{MnBr}_4]^{2-}$ is 5.9 BM

In the light of above statements, choose the correct answer from the options given below

- (1) Both statement I and statement II are correct
 (2) Both statement I and statement II are incorrect
 (3) Statement I is correct but statement II is incorrect
 (4) Statement I is incorrect but statement II is correct

39. An aromatic compound 'P' on treatment with aqueous ammonia and heating forms compound 'Q' which on treatment with LiAlH_4 followed by hydrolysis forms a compound 'R' of molecular formula $\text{C}_7\text{H}_9\text{N}$. The incorrect statement about R is

- (1) It gives carbylamine reaction
- (2) It does not react with Hinsberg's reagent
- (3) It is more basic than N-methylaniline
- (4) It reacts with benzoyl chloride to form N-benzylbenzamide

40. Consider the following statements

- (a) Except glycine, all other naturally occurring α -amino acids are optically active
 - (b) Methionine contains sulphur atom
 - (c) Threonine is a non-essential amino acid.
- The correct statements are

- (1) (a) and (b) only
- (2) (b) and (c) only
- (3) (a) and (c) only
- (4) (a), (b) and (c)

41. The maximum number of electrons in an atom that can have quantum number values $n = 3, l = 1$ are

- (1) 6
- (2) 2
- (3) 10
- (4) 18

42. Which among the following is not a group IV cation?

- (1) Ni^{2+}
- (2) Co^{2+}
- (3) Zn^{2+}
- (4) Ba^{2+}

43. Given below are two statements.

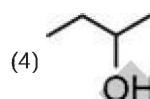
Statement I: MnO_4^{2-} undergoes disproportionation in acidic medium.

Statement II: V_2O_3 is basic and V_2O_5 is amphoteric.

In the light of above statements, choose the correct answer from the options given below

- (1) Statement I is incorrect and statement II is correct.
- (2) Both statement I and statement II are correct.
- (3) Both statement I and statement II are incorrect.
- (4) Statement I is correct and statement II is incorrect.

44. Which alcohol on reaction with Cu at 573 K gives alkene as a major product?



45. The maximum number of moles of KMnO_4 needed in acidic medium to react with the same moles of the following compounds, will be:

- (1) FeSO_4
- (2) FeC_2O_4
- (3) KI
- (4) $\text{Fe}_2(\text{C}_2\text{O}_4)_3$

PHYSICS

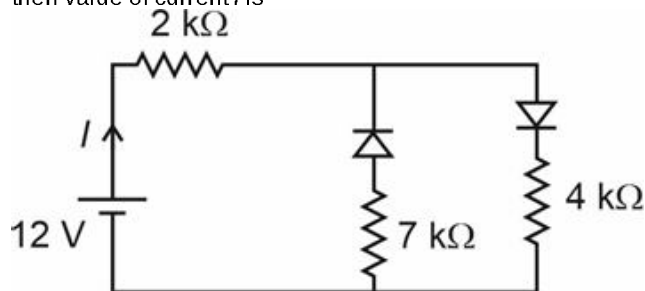
46. The vernier scale of a callipers is divided into 20 divisions which coincide with 18 main scale divisions. Each main scale division is 0.2 mm. The least count of the instrument is

- (1) 0.2 mm
- (2) 0.02 mm
- (3) 0.1 mm
- (4) 0.01 mm

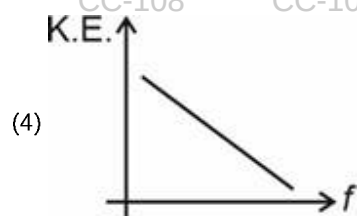
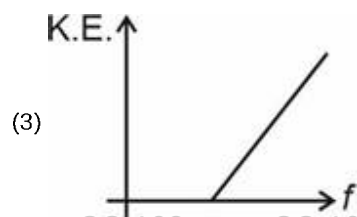
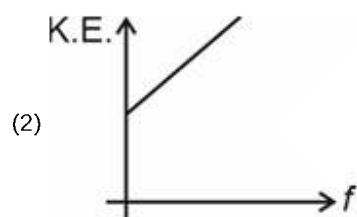
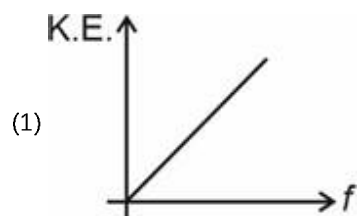
47. If the energy of a hydrogen atom in n^{th} orbit is E , then energy in the n^{th} orbit of a singly ionized helium atom will be

- (1) $\frac{E}{4}$
- (2) $4E$
- (3) $2E$
- (4) $\frac{E}{2}$

48. If diode connected in diagram are considered to be ideal, then value of current I is



- (1) 1 mA
 (2) $\frac{4}{3}$ mA
 (3) 2 mA
 (4) $\frac{3}{4}$ mA
49. According to Einstein's photoelectric equation, the graph between the maximum kinetic energy (K.E.) of photoelectrons ejected and the frequency (f) of incident radiation is best represented by



50. The electrical conductivity of a semiconductor increases when electromagnetic radiation of wavelength shorter than $6200 \text{ \AA}$, is incident on it. The band gap in (eV) for the semiconductor is

- (1) 1
 (2) 2
 (3) 0.7
 (4) 1.1

51. Surface area of the nucleus (X^8) is A_0 , then surface area of nucleus (Y^{64}) will be

- (1) A_0
 (2) $4A_0$
 (3) $\frac{A_0}{2}$
 (4) $2A_0$

52. Consider the following statements

- (i) An intrinsic semiconductor will behave as insulator at $T = 0 \text{ K}$.
 (ii) Doping pure silicon with trivalent impurities gives p-type semiconductors.
 (iii) The majority carriers in n-type semiconductors are electrons.
 (iv) Solar cell works when it is in forward bias

The correct statement(s) is/are

- (1) Only (i) and (ii)
 (2) Only (i) and (iii)
 (3) Only (i), (ii) and (iii)
 (4) Only (i), (ii) and (iv)

53. The electric field in a region is directed outward and is proportional to the distance r from the origin. If we take a spherical volume of radius r taking centre at the origin, then the charge contained inside the volume is proportional to

- (1) r
 (2) r^2
 (3) r^0
 (4) r^3

54. A wire of resistance 12Ω is bent to form a circle. The equivalent resistance between two diametrically opposite points will be

- (1) 3Ω
 (2) 6Ω
 (3) 12Ω
 (4) 18Ω

55. Consider the following statements and choose the correct option

Statement (A): Electromotive force (emf) is not a force, it is the voltage difference between two terminal of a battery in open circuit.

Statement (B): The potential difference across terminals of a battery having some internal resistance can never be greater than the emf of the battery.

- (1) Statement (A) is correct while statement (B) is incorrect
- (2) Statement (A) is incorrect while statement (B) is correct
- (3) Both the statements are correct
- (4) Both the statements are incorrect

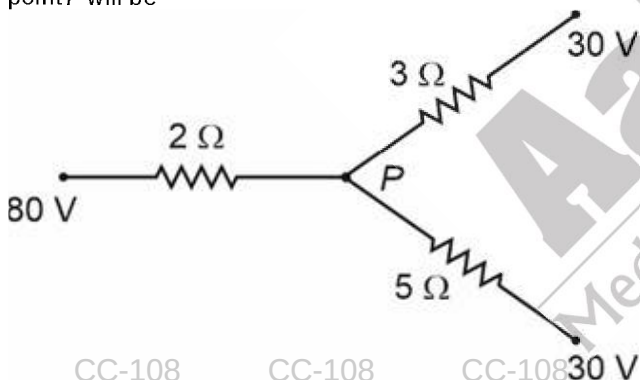
56. Two identical capacitors each of capacitance C and breakdown voltage V are connected in series. The respective values of capacitance and the breakdown voltage of the combination will be

- (1) C, V
- (2) $\frac{C}{2}, 2V$
- (3) $C, 2V$
- (4) $2C, 2V$

57. A neutral body is charged positively by rubbing it, its weight

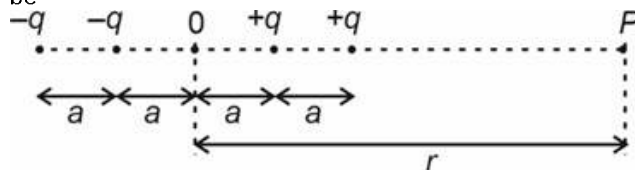
- (1) Increases slightly
- (2) Decreases slightly
- (3) Remains constant
- (4) May increase or decrease

58. A part of an electric circuit is given below. The potential at point P will be



- (1) $\frac{100}{31} \text{ V}$
- (2) $\frac{200}{31} \text{ V}$
- (3) $\frac{1680}{31} \text{ V}$
- (4) $\frac{700}{31} \text{ V}$

59. Four point charges are placed on a line as shown. If $r \gg a$, then potential at point P due to this system of charges will be

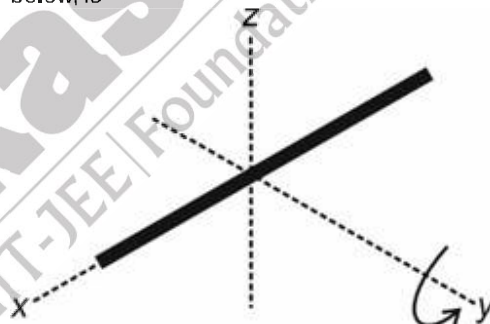


- (1) $\frac{6qa}{\pi\epsilon_0 r^2}$
- (2) $\frac{3qa}{2\pi\epsilon_0 r^2}$
- (3) $\frac{9qa}{2\pi\epsilon_0 r^2}$
- (4) $\frac{qa}{2\pi\epsilon_0 r^2}$

60. Critical angle of light passing from water to air is maximum for

- (1) Yellow
- (2) Green
- (3) Blue
- (4) Indigo

61. Moment of inertia of a thin rod of mass M and length L , about y -axis (passing through mid-point of rod) as shown below, is



- (1) $\frac{ML^2}{24}$
- (2) $\frac{ML^2}{12}$
- (3) $\frac{ML^2}{6}$
- (4) $\frac{ML^2}{3}$

62. A ray of light is incident normally on one face of a thin prism of refractive index μ and emerges at an angle (e) from the normal. Angle of prism A is nearly

- (1) $\frac{e}{\mu}$
- (2) μe
- (3) $\frac{\pi - e}{\mu}$
- (4) $\mu \left(\frac{\pi}{2} - e \right)$

63. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).

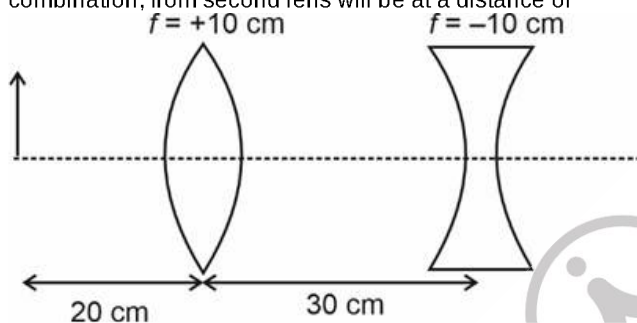
Assertion (A): Angular momentum conservation can be used to explain Kepler's second law of planetary motion.

Reason (R): Areal velocity of a planet revolving around the sun is equal to its angular momentum.

In the light of the above statements, choose the correct answer from the options given below:

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A)
- (2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
- (3) (A) is true but (R) is false
- (4) Both (A) and (R) are false

64. The position of final image, formed by given lens combination, from second lens will be at a distance of



- (1) 5 cm
- (2) 10 cm
- (3) 20 cm
- (4) Infinity

65. The fractional change in acceleration due to gravity on surface of earth (g) when the body is taken to a depth h from the surface of earth, is (where R = radius of Earth)

- (1) $1 - \frac{h}{R}$
- (2) $\frac{h}{R}$
- (3) $1 - \frac{2h}{R}$
- (4) $\frac{2h}{R}$

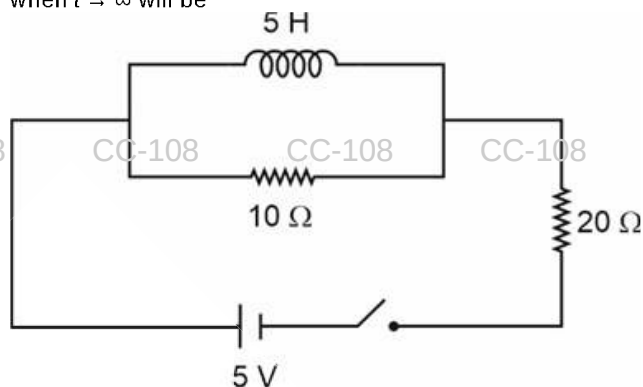
66. The centre of mass of a system of particles depends on

- (1) Position of the particles
- (2) Relative distance between the particles
- (3) Masses of the particles
- (4) All of these

67. Power consumed in A.C circuit is zero then the ac source could be connected to

- (1) Resistor only
- (2) Inductor only
- (3) Capacitor only
- (4) Both (2) and (3)

68. Consider a network of resistors and inductor as shown. The key is closed at $t = 0$, then the current through the battery when $t \rightarrow \infty$ will be



- (1) $\frac{1}{2}$ A
- (2) $\frac{2}{3}$ A
- (3) $\frac{1}{4}$ A
- (4) Zero

69. Speed of EM wave in free space is given by (where symbols have their usual meaning)

- (1) $c = \sqrt{\frac{\epsilon_0}{\mu_0}}$
- (2) $c = \sqrt{\mu_0 \epsilon_0}$
- (3) $c = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$
- (4) $c = \sqrt{\frac{\mu_0}{\epsilon_0}}$

70. Consider two long solenoids A and B having length $2L$ and $3L$ and number of loop as N and $2N$ respectively. If both have same current then ratio of magnetic field inside A to that of the B will be

- (1) $\frac{3}{4}$
- (2) $\frac{3}{2}$
- (3) 1
- (4) $\frac{1}{2}$

71. A bar magnet of length l and pole strength m is placed in uniform magnetic field B at an angle of 60° with field. The torque on the bar magnet at this instant will be

- (1) $\frac{mBl}{2}$
- (2) $\frac{\sqrt{3}mBl}{2}$
- (3) mBl
- (4) $2mBl$

72. If potential difference between point A and B ($V_A - V_B$) of a given part of a circuit as shown in the figure is 4 V. Then current in circuit will



- (1) Increasing at the rate of 1000 A/s
- (2) Increasing at the rate of 2000 A/s
- (3) Decreasing at the rate of 1000 A/s
- (4) Decreasing at the rate of 2000 A/s

73. Consider an a.c circuit having resistor of resistance R and an inductor having reactance X_L . If voltage leads the current by an angle 60° , then

- (1) $X_L = R$
- (2) $X_L = 2R$
- (3) $X_L = \sqrt{3}R$
- (4) $X_L = \sqrt{2}R$

74. Two quantities are measured as $P = (1 \pm 0.40) \text{ m}$, $Q = (4 \pm 0.20) \text{ m}$. The correct value of $(PQ)^{1/2}$ will be

- (1) $(4 \pm 0.09) \text{ m}$
- (2) $(2 \pm 0.01) \text{ m}$
- (3) $(2 \pm 0.45) \text{ m}$
- (4) $(4 \pm 0.01) \text{ m}$

75. A particle begins to move along straight line where the acceleration (a) of the particle varies with displacement (x) according to relation, $a = 5x$, then velocity of the particle varies with displacement as

- (1) $x^{1/2}$
- (2) x^1
- (3) $x^{1/3}$
- (4) $x^{3/4}$

76. Thin copper wire of length L increases in length by 2% when heated from T_1 to T_2 . If a copper cube having side $10L$ is heated from T_1 to T_2 then the percentage change in volume of the cube is

- (1) 2%
- (2) 3%
- (3) 6%
- (4) 8%

77. If β_1 , β_2 and β_3 are fringe widths for red, green and violet light respectively for the same setup of Young's double slit experiment, then

- (1) $\beta_1 = \beta_2 = \beta_3$
- (2) $\beta_1 > \beta_3 > \beta_2$
- (3) $\beta_2 > \beta_1 > \beta_3$
- (4) $\beta_1 > \beta_2 > \beta_3$

78. Consider the given statements and choose the correct option that follows:

Statement 1: During a collision the total linear momentum of system is conserved at each instant of collision.

Statement 2: During a collision the kinetic energy conservation holds always.

Based on above information, pick the correct option.

- (1) Both statements (1) and (2) are true
- (2) Both statements (1) and (2) are false
- (3) Statement (1) is true but (2) is false
- (4) Statement (1) is false but (2) is true

79. The energy used by a 100 W bulb which is on for 10 hours is

- (1) 1 kWh
- (2) $3.6 \times 10^6 \text{ J}$
- (3) $4.8 \times 10^5 \text{ J}$
- (4) Both (1) and (2)

80. A car starts from rest, accelerates uniformly at 2 m/s^2 . The distance travelled by the car in fourth second is

- (1) 2.25 m
- (2) 14 m
- (3) 7 m
- (4) Zero

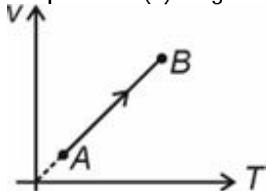
81. If maximum speed and maximum acceleration of a particle executing SHM is found to be 5 cm/s and $50\pi \text{ cm/s}^2$ respectively, then its time period will be

(1) 5 s
 (2) 2 s
 (3) $\frac{1}{5} \text{ s}$
 (4) $\frac{1}{10} \text{ s}$

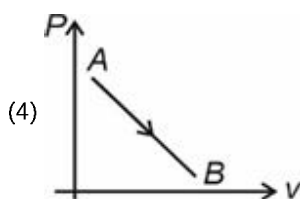
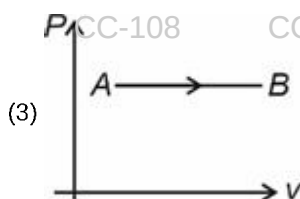
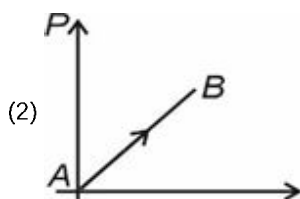
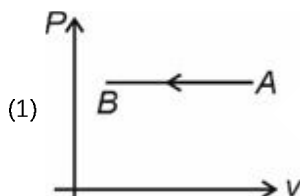
82. A gas mixture consists of 5 moles of oxygen and 3 moles of argon at temperature T . Assuming the gases to be ideal and oxygen bond to be rigid, the total internal energy of the mixture is (where R denotes the universal gas constant)

(1) $17RT$
 (2) $15RT$
 (3) $20RT$
 (4) $11RT$

83. A thermodynamic process is shown on a volume (V) – temperature (T) diagram.



The correct representation of above process on P - V diagram will be



84. Consider the following statements

(A) Temperature of a body is related to its average internal energy, not to the kinetic energy of motion of its centre of mass.

(B) The first law of thermodynamics is based on law of conservation of energy applied to any system in which energy transfer from or to the surrounding (through heat and work) is taken into account.

Based on above information, pick correct option

(1) Both statements (A) and (B) are true
 (2) Both statements (A) and (B) are false
 (3) Statement (A) is true while (B) is false
 (4) Statement (B) is true while (A) is false

85. Consider the following statements

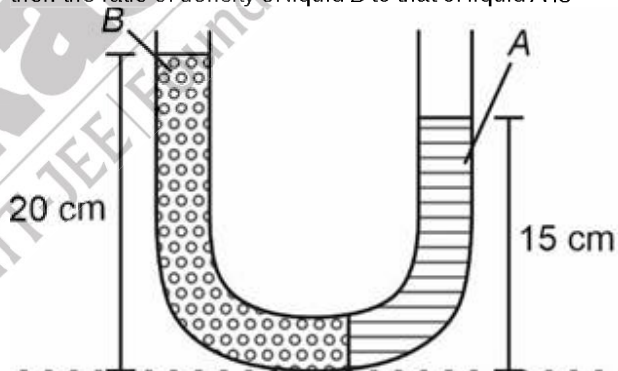
(A) Convection involves flow of matter within a fluid due to unequal temperatures of its parts.

(B) The change of solid state to vapour state without passing through liquid state is called sublimation.

Based on above information, pick correct option

(1) Both statements are correct
 (2) Both statements are incorrect
 (3) Only statement A is correct
 (4) Only statement B is correct

86. In a U-tube as shown in figure, liquid A is on the right side of tube while liquid B is on the left side. The heights from the bottom for liquid A and B is 15 cm and 20 cm respectively, then the ratio of density of liquid B to that of liquid A is



(1) $\frac{3}{4}$
 (2) $\frac{4}{3}$
 (3) $\frac{5}{7}$
 (4) $\frac{7}{5}$

87. Match List-I with List-II where list-I denotes nature and medium of wave and list-II denotes the expression for speed of wave.

(All symbols have their usual meaning)

List-I	List-II
a. Transverse wave on a stretched string	(i) $\sqrt{\frac{Y}{\rho}}$
b. Longitudinal wave in a metallic bar	(ii) $\sqrt{\frac{E}{\rho}}$
c. Longitudinal wave in a fluid	(iii) $\sqrt{\frac{T}{\mu}}$

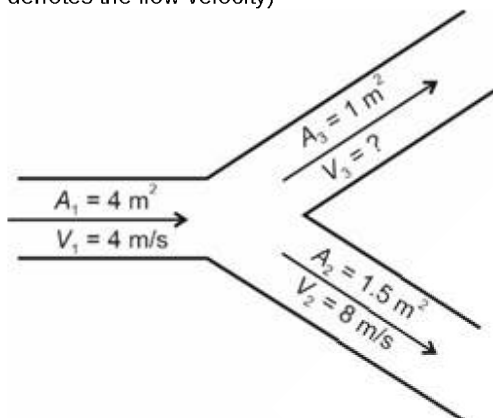
(1) a(ii), b(iii), c(i)

(2) a(i), b(ii), c(iii)

(3) a(iii), b(ii), c(i)

(4) a(iii), b(i), c(ii)

88. An ideal incompressible liquid flows through a tube as shown. (where A denotes the area of cross section and v denotes the flow velocity)



Based on above information, velocity v_3 will be equal to

(1) 4 m/s

(2) 8 m/s

(3) 12 m/s

(4) 16 m/s

89. A particle is projected with a speed of 20 m s^{-1} from level ground at an angle θ equal to 45° from horizontal. The ratio of maximum height attained by the body to horizontal range acquired by the body will be

(1) $\frac{2}{1}$

(2) $\frac{1}{2}$

(3) $\frac{1}{4}$

(4) $\frac{3}{2}$

90. Given below are two statements: one is labelled as **Assertion A** and the other is labelled as **Reason R**.

Assertion (A): The unit of stress is same as that of pressure.

Reason (R): Stress is a vector quantity.

In the light of above statements, select the correct option.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) Both (A) and (R) are false

BIOLOGY

91. Which of the following statements is **incorrect** w.r.t. rules of binomial nomenclature?

The first component in a biological name represents its

(1) genus while the second component denotes the specific epithet.

(2) The first word denoting genus starts with a capital letter, while the specific epithet starts with a small letter.

(3) The name of the author or discoverer is written after specific epithet in abbreviated form.

(4) Biological names are generally taken from Greek language irrespective of their origin

92. Select the **incorrectly** matched pair

(1) <i>Puccinia</i>	–	Causes rust disease in wheat
(2) <i>Ustilago</i>	–	Causes smut disease
(3) <i>Claviceps</i>	–	Causes ergot disease
(4) <i>Aspergillus</i>	–	Causes white rust on mustard leaves

(1) (1)

(2) (2)

(3) (3)

(4) (4)

93. Sleeping sickness in humans is caused by a/an

- (1) Amoeboid protozoan
- (2) Flagellated protozoan
- (3) Ciliated protozoan
- (4) Sporozoan

94. Match column I with column II and select the correct option

Column I	Column II
a. Opposite phyllotaxy	(i) Citrus
b. Alternate phyllotaxy	(ii) China rose
c. Monoadelphous	(iii) Mustard
d. Polyadelphous	(iv) <i>Calotropis</i>

- (1) a(iv), b(iii), c(ii), d(i)
- (2) a(i), b(ii), c(iv), d(iii)
- (3) a(iii), b(iv), c(i), d(ii)
- (4) a(ii), b(iv), c(i), d(iii)

95. In which of the following plants the ovules are borne on central axis and septa are absent in the ovary?

- (1) Sunflower
- (2) Marigold
- (3) Mustard
- (4) *Dianthus*

96. Read the following statements (A-D)

- A. *Gonyaulax* undergo rapid multiplication and they make the sea appear red.
 B. All protozoans are heterotrophs and live as predators or parasites.
 C. Dinoflagellates have a protein rich layer called pellicle which makes their body flexible.
 D. Diatoms are the chief producers in the oceans.
 Select the correct set of statements from the options given below.

- (1) Only A and C
- (2) Only A, B and D
- (3) Only B and D
- (4) Only B and C

97. Select odd one out w.r.t. zygomorphic flower.

- (1) Pea
- (2) *Datura*
- (3) Gulmohur
- (4) *Cassia*

98. Consider the following statements and select the correct option.

Statement A: The NADP reductase enzyme is located on the lumen side of thylakoid membrane.

Statement B: Splitting of water molecules occurs towards the inner side of the thylakoid membrane.

- (1) Only statement A is correct
- (2) Only statement B is correct
- (3) Both the statements A and B are correct
- (4) Both the statements A and B are incorrect

99. Cyclic photophosphorylation differs from non-cyclic photophosphorylation as the former

- (1) Involves release of ATP, NADPH and O₂
- (2) Occurs in the lamellae of the grana
- (3) Involves only the photosystem having reaction centre with absorption maxima at 700 nm.
- (4) Involves both PS I and PS II

100. Read the following statements and choose the correct option.

Statement I: Temperature regulates the decomposition of detritus by affecting activities of soil microbes.

Statement II: Earthworm helps in the breakdown of complex organic matter as well as loosening of soil.

- (1) Only statement II is true
- (2) Both statements I and II are true.
- (3) Only statement I is true
- (4) Both statements I and II are false

101. Match column I with column II and choose the correct option

Column I	Column II
a. Complex I	(i) Cytochrome bc ₁ complex
b. Complex III	(ii) NADH dehydrogenase
c. Complex II	(iii) Cytochrome c oxidase complex
d. Complex IV	(iv) Succinate dehydrogenase

- (1) a(iii), b(i), c(iv), d(ii)
- (2) a(ii), b(iv), c(i), d(iii)
- (3) a(ii), b(i), c(iv), d(iii)
- (4) a(iii), b(iv), c(i), d(ii)

102. Choose the correct option for net primary productivity

- (1) It is expressed in terms of weight as (kcal m⁻²)
- (2) It is the rate of formation of new organic matter by consumers
- (3) It is the available biomass for consumption to heterotrophs and decomposers
- (4) It is the amount of the formation of organic matter during photosynthesis.

103. The first irreversible reaction of glycolysis is catalysed by

- (1) Aldolase
- (2) Invertase
- (3) Phosphofructokinase
- (4) Hexokinase

104. The earliest system of classification of organisms

- (1) Used only gross superficial morphological characters
- (2) Were based on natural affinities among the organisms
- (3) Considered both external and internal features
- (4) Were based on evolutionary relationships and cytological information.

105. Some members of which division of the plant kingdom show the event that is precursor to the seed habit?

- (1) Algae
- (2) Pteridophytes
- (3) Bryophytes
- (4) Gymnosperms

106. To find out the type of organic compounds found in living organisms, a chemical analysis is carried out for the living tissue, in which it is grinded with a chemical named

- (1) Trihydroxypropane
- (2) Trichloroacetic acid
- (3) Arachidonic acid
- (4) Guanylic acid

107. Lysine can be differentiated from valine as

- (1) Former possesses more number of carboxyl group
- (2) Latter possesses equal number of amino and carboxyl groups
- (3) Latter possesses a cyclic ring in its structure
- (4) Former is acidic and latter is basic in chemical nature

108. Glycogen is similar to starch as both of them

- (1) Produce blue colour with iodine
- (2) Lack complex helices
- (3) Are polymers of glucose monomers
- (4) Are present as a store house of energy in plant tissues

109. Match column I with column II and select the **correct** option.

	Column I		Column II
a.	Collagen	(i)	Most abundant protein in the whole of the biosphere
b.	RuBisCO	(ii)	Most abundant protein in animal world
c.	Cellulose	(iii)	Hormone that increases blood glucose levels
d.	Glucagon	(iv)	Homopolymer which is also a secondary metabolite

- (1) a(ii), b(iv), c(iii), d(i)
- (2) a(iii), b(iv), c(ii), d(i)
- (3) a(i), b(ii), c(iii), d(iv)
- (4) a(ii), b(i), c(iv), d(iii)

110. The type of epithelium lining the stomach, serves mainly the function of

- (1) Secretion and absorption
- (2) Forming a diffusion boundary for exchange of gas
- (3) Moving particles or mucus in a specific direction over the epithelium
- (4) Providing protection against chemical and mechanical stresses

111. Select the **incorrect** option w.r.t cerebellum of humans.

- (1) Has very convoluted surface to accommodate more neurons
- (2) Integrates information received from the semicircular canals of the ear and the auditory system
- (3) Located below the brain stem
- (4) Largest part of the hindbrain

112. The construction of the first recombinant DNA by Stanley Cohen and Herbert Boyer was done in the year

- (1) 1980
- (2) 1971
- (3) 1982
- (4) 1972

113. In *Hind* III, the letter 'd' indicates the _____ from which it has been isolated.

Select the **correct** option to fill in the blank.

- (1) Strain of the eukaryotic cell
- (2) Species of the prokaryotic cell
- (3) Genus of the eukaryotic cell
- (4) Strain of the prokaryotic cell

114. All of the following are true w.r.t *Taq* polymerase, **except**

- (1) Isolated from bacterium
- (2) Thermolabile in nature
- (3) Used in extension step of PCR
- (4) Is a DNA dependent DNA polymerase

115. If a gene encoding the enzyme β -galactosidase is inserted within the selectable marker gene of the pBR322 then after transformation in the presence of a chromogenic substrate

- (1) Non-transformants will produce blue coloured colonies
- (2) All transformants will produce white coloured colonies due to inactivation of selectable marker gene
- (3) All recombinants will produce blue coloured colonies
- (4) Non-recombinant transformants will produce blue coloured colonies as they carry the plasmid

116. Which of the following statements is **false**?

- (1) The same yeast used for bread-making, is also used for fermenting malted cereals to produce ethanol
- (2) The large holes in 'Swiss cheese' are due to the production of a large amount of CO_2 by a fungus
- (3) The dough which is used for making foods such as Dosa and Idli is fermented by using bacteria
- (4) *Lactobacillus*, a LAB, play a very beneficial role in checking disease causing microbes in our stomach

117. Read the following Assertion (A) and Reason (R) statements and select the **correct** option

Assertion (A): A population has certain attributes that an individual does not.

Reason (R): An individual may have birth rates and death rates, but a population has birth and death.

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A)
- (2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
- (3) Both (A) and (R) are false
- (4) (A) is true but (R) is false

118. Identify the **incorrectly** matched pair.

(1) Over-exploitation	— Stellar's sea cow and passenger pigeon
(2) Evil quartet	— Habitat loss and fragmentation
(3) Co-extinction	— Fig and fig wasp
(4) Nile perch introduced in Lake Victoria	— Extinction of <i>Clarias gariepinus</i>

- (1) (1)
- (2) (2)
- (3) (3)
- (4) (4)

119. Identify the statements which are **true** for both brown and red algae.

- a. Sexual reproduction is anisogamous.
- b. They are found primarily in marine habitats.
- c. Vegetative reproduction takes place by fragmentation.
- d. Gametes are pyriform and bear two laterally attached flagella.
- e. Cell wall is composed of cellulose and hydrocolloids.

In the light of the above statements, select the **correct** option

- (1) b and d
- (2) a and e
- (3) b, c and e
- (4) a, c and d

120. Which among the following fermented beverages has lowest ethanol concentration?

- (1) Whisky
- (2) Brandy
- (3) Rum
- (4) Wine

121. Human population is said to be mature or stable when

- (1) Reproductive individuals are more than pre-reproductive individuals
- (2) Reproductive individuals are less than post-reproductive individuals
- (3) Pre-reproductive individuals are less than post-reproductive individuals
- (4) Pre-reproductive individuals being only marginally more than the reproductive individuals

122. The species which is confined to a particular region and not found anywhere else is known as

- (1) Keystone species
- (2) Threatened species
- (3) Endemic species
- (4) Vulnerable species

123. The growth in plants

- a. Is generally indeterminate
 - b. Is a closed form of growth
 - c. Is due to the presence of meristematic cells
- The **correct** one(s) is/are

- (1) a and b only
- (2) b and c only
- (3) a and c only
- (4) a only

124. Select the phytohormone that helps in increasing the yield of sugarcane by increasing the length of its stem.

- (1) Ethylene
- (2) Auxin
- (3) Gibberellin
- (4) Absciscic acid

125. Select the **odd** one w.r.t. pollen grain in angiosperms.

- (1) Generally spherical, about 25-50 μm in diameter
- (2) Inner layer has prominent apertures called germ pores
- (3) Exine can withstand high temperatures and strong acids
- (4) Outer layer is made up of one of the most resistant organic materials

126. Multicarpellary, syncarpous gynoecium can be seen in

- a. *Michelia*
- b. *Papaver*
- c. *Hibiscus*

- (1) a and b
- (2) b and c
- (3) a and c
- (4) c only

127. Which of the following statements is **incorrect**?

- (1) Synergids secrete some chemotropic substance to direct the growth of pollen tube
The most common type of embryo sac in angiosperm is
- (2) monosporic which is formed by three sequential mitotic divisions.
- (3) Antipodal cells are the vegetative cells of embryo sac
In angiosperms, the egg apparatus of embryo sac
- (4) consists of one egg cell, two synergids, three antipodal cells and two polar nuclei

128. In which of the following genetic disorders, a single protein that is a part of the cascade of proteins involved in blood clotting is affected?

- (1) Sickle-cell anaemia
- (2) Haemophilia
- (3) Phenylketonuria
- (4) Thalassemia

129. β -Thalassemia in human beings is controlled by

- (1) Two closely linked genes HBA1 and HBA2
- (2) A single gene HBB on chromosome 11
- (3) A single pair of alleles, Hb^A and Hb^S
- (4) Chromosome number 21

130. Select one of the reasons that is true for choosing *Drosophila melanogaster* as a model organism for the experimental verification of chromosomal theory of inheritance, by T.H. Morgan.

- (1) A single mating produced very few number of offsprings
- (2) They have short life cycle of about two weeks
- (3) Females are not easily distinguishable from males
- (4) They could be grown on natural medium only

131. A coelomate non-vertebrate chordate among the following in which notochord is persistent throughout the life is

- (1) *Petromyzon*
- (2) *Branchiostoma*
- (3) *Scoliodon*
- (4) *Trygon*

132. Which of the following statements is not **incorrect** w.r.t. kidneys of a healthy adult human?

- (1) Bean shaped structures situated between the levels of 7th thoracic and first lumbar vertebrae.
- (2) The cumulative weight of both the kidneys is 180-270 gms.
- (3) Towards the centre of outer convex surface hilum is present.
- (4) Ureter, nerves and blood vessels enter in the kidneys through hilum.

133. Which of the following is one of the connotation of theory of special creation?

- (1) Life appeared 2000 mya
- (2) The earth is about 4000 years old
- (3) Life comes only from pre-existing life
- (4) All life forms evolve in water only

134. The Big Bang theory attempts to explain to us the origin of

- (1) Milky way galaxy only
- (2) Earth only
- (3) Universe
- (4) Solar system only

135. Listed below are some facts w.r.t. evolution.

- Hugo deVries stated that any population has built in variation in characteristics.
- According to Darwin, the fitness refers ultimately and only to reproductive fitness.
- Brain of fishes and that of humans are homologous structures.
- Heart of amphibians and cyclostomes are analogous structures.

Which of the above stated facts are **not** true?

- a and b
- b and c
- a and c
- a and d

136. Match column I with column II and select the **correct** option.

	Column I		Column II
a.	Crustaceans	(i)	Malpighian tubules
b.	Insects	(ii)	Nephridia
c.	Earthworms	(iii)	Protonephridia
d.	Cephalochordates	(iv)	Antennal glands

- a(i), b(ii), c(iii), d(iv)
- a(ii), b(iii), c(i), d(iv)
- a(iii), b(i), c(iv), d(ii)
- a(iv), b(i), c(ii), d(iii)

137. Complete the analogy by selecting the **correct** option.

Psilophyton: Conifers :: _____ : Seed ferns

- Gnetales
- Ginkgos
- Cycads
- Arborescent lycopods

138. Islets of Langerhans in a normal human pancreas represent only _____ of the pancreatic tissue. Choose the correct option to fill the blank.

- 89-90 per cent
- 1-2 per cent
- 98-99 per cent
- 10-11 per cent

139. Consider the following features

- Vector borne disease
- Excess mucus and blood clots in stools
- Caused by protozoan
- Spread through faeco-oral route

Which of the above features is/are **true** for both malaria and amoebic dysentery?

- a and d
- Only c
- Only d
- b, c and d

140. A hormone 'X' is a secretion of the gland located at the back side of thyroxine synthesising gland. Which of the following options is correct about 'X'?

- It supports erythropoiesis
- It can stimulate the process of bone dissolution and demineralisation
- It plays a major role in differentiation of T-lymphocytes
- It is a steroidal hormone

141. How many intrachain disulphide bonds are present in light chains of a monomeric antibody molecule?

- Four
- Six
- Eight
- Ten

142. Consider the given statements w.r.t human female mammary glands.

Statement A: The glandular tissue of each breast is divided into 30-40 mammary lobes containing clusters of alveoli.

Statement B: Several mammary ducts join to form a wider mammary ampulla which is connected to mammary tubule through which milk is sucked out.

Select the **correct** option.

- Both statements A and B are correct
- Both statements A and B are incorrect
- Only statement A is correct
- Only statement B is correct

143. The cyclical changes in the ovary and the uterus during menstrual cycle are induced by changes in the levels of

- Pituitary hormones only
- Ovarian hormones only
- Hypothalamic hormones only
- Both ovarian and pituitary hormones

144. In humans, sex of embryo is determined by the presence of

- X chromosome in ootid only
- Presence of X chromosome in sperm only
- Presence of both X and Y chromosomes in the sperm
- Presence of either X or Y chromosome in the sperm

145. The body of cockroach is segmented and divisible into three distinct regions called

- Head, neck and abdomen
- Head, thorax and abdomen
- Cephalothorax, abdomen and tail
- Head, trunk and tail

146.Assertion (A) : During pregnancy, the levels of hormones such as GH, thyroxine, gonadotropins, estrogen and progesterone are increased several folds in blood.

Reason (R): Increased production of all these hormones is essential for supporting the fetal growth, metabolic changes in the mother and maintenance of pregnancy.

In the light of above statements, choose the **correct** answer from the options given below.

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A)
- (2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
- (3) (A) is true but (R) is false
- (4) Both (A) and (R) are false

147.Choose the **incorrect** match w.r.t. frog.

(1)	Skin	-	Highly vascularised
(2)	Rectum	-	Opens into cloaca
(3)	Pancreas	-	One of the main digestive glands
(4)	Fertilization	-	Internal

- (1) (1)
- (2) (2)
- (3) (3)
- (4) (4)

148.The whole plant can be regenerated from any part of a plant, when grown in a test tube under sterile conditions in special nutrient media. This nutrient medium must contain all of the following **except**,

- (1) A carbon source
- (2) Growth regulators
- (3) Vitamins
- (4) Enzymes

149.*Bacillus thuringiensis* produces a toxin protein which exists as inactive protoxins, implies that

- (1) This toxin do not affect the insect until and unless it is broken down by digestive enzyme in hindgut.
- (2) It becomes activated in oral cavity as soon as insect consumes this.
- (3) These toxins do not affect the *Bacillus* itself
- (4) These toxins only get activated in *Bacillus* in acidic pH

150.During cardiac cycle two prominent sounds are produced in a healthy human's heart. Which of the following is correct about these sounds?

- (1) 'Dub' is associated with the closure of the tricuspid and bicuspid valves
- (2) 'Lub' is associated with the closure of semilunar valves
- (3) Second heart sound 'dub' is produced during atrial systole
- (4) First heart sound 'lub' is produced during ventricular systole

151.In human heart, Purkinje fibres are specialised

- (1) Epithelial tissue
- (2) Connective tissue
- (3) Muscular tissue
- (4) Nervous tissue

152.Match column I with column II

	Column I		Column II
a.	Red muscle fibre	(i)	Tail part of meromyosin
b.	White muscle fibre	(ii)	Globular head with short arm
c.	Heavy meromyosin (HMM)	(iii)	High amount of sarcoplasmic reticulum
d.	Light meromyosin (LMM)	(iv)	Plenty of mitochondria

Select the **correct** option

- (1) a(iv), b(iii), c(ii), d(i)
- (2) a(i), b(ii), c(iii), d(iv)
- (3) a(iv), b(iii), c(i), d(ii)
- (4) a(iv), b(i), c(ii), d(iii)

153.Assertion (A) : During dissection and study of an adult *Hirudinaria*, both ovaries and testes were found in the same segmented worm.

Reason (R): Leeches are dioecious and perform sexual reproduction.

In the light of above statements, choose the **correct** option

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A)
- (2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
- (3) (A) is true but (R) is false
- (4) Both (A) and (R) are false

154.Commercial availability and production of recombinant form of insulin was possible due to

- a. Combination of A and C chains by creating a type of bond which is also present in an antibody molecule.
- b. Production of chains A, B and C together in *E.coli*.
- c. Success in getting insulin assembled into a mature form.

Select the **correct** option.

- (1) Only a
- (2) Only b and c
- (3) Only c
- (4) a, b and c

155. Read the following statements

- If heart beats 100 times in a minute, the duration of cardiac cycle will be 0.9 sec.
- The atrium and ventricle of the same side are separated by thin muscular tissue in human heart called atrio-ventricular septum.
- A thin muscular wall called the inter-atrial septum separates the right and left atria in human heart.

Choose the option with only **correct** statement(s).

- a, b and c
- Only c
- Only a
- b and c

156. Read the statements given below

Statement I: Morgan conducted dihybrid crosses using *Drosophila* and found out that the proportion of parental gene combination was much lower than the non-parental gene combination.

Statement II: Linkage describes the physical association of genes on a chromosome whereas, recombination describes the generation of non-parental gene combinations.

Choose the **correct** option for the above given statements.

- Statement I is true and statement II is false
- Statement II is true and statement I is false
- Both the statements I and II are true
- Both the statements I and II are false

157. Conjoint and closed type of vascular bundles are found in all the given plant parts, **except**

- Monocot leaf
- Dicot stem
- Monocot stem
- Dicot leaf

158. Read the following statements and choose the option for **incorrect** one(s).

- In leaves of grasses, abaxial epidermal cells along the veins modify themselves into the cells which can minimise water loss during conditions of water stress.
- In sunflower root, large and well developed pith is seen in its transverse section.
- During elongation of stem, some cells, left behind from shoot apical meristem constitute the axillary bud.
- Chlorenchyma is a chloroplast containing collenchyma which performs photosynthesis.
- Protoxylem consists of narrow vessels while metaxylem has broader ones.

- Only A, C and E
- Only A
- All except C
- Only A, B and D

159. Which amongst the following have RNA as the genetic material?

- Yeast and bacteria
- E. coli* and Human
- Tobacco mosaic virus and QB bacteriophage
- $\phi \times 174$ bacteriophage and *E. coli*

160. Match List I with List II

	List I		List II
a.	Nirenberg and Matthaei	(i)	Performed base composition studies for dsDNA
b.	Taylor <i>et al.</i>	(ii)	Performed series of experiments by selecting two strains of <i>Streptococcus</i>
c.	Erwin Chargaff	(iii)	Used a synthetic poly U RNA and deciphered the code by translating this as polyphenylalanine
d.	Frederick Griffith	(iv)	Proved that chromosomes replicate semi-conservatively in eukaryotes

Choose the **correct** option

- a(iii), b(ii), c(iv), d(i)
- a(ii), b(i), c(iii), d(iv)
- a(i), b(iii), c(ii), d(iv)
- a(iii), b(iv), c(i), d(ii)

161. The presence of A is reminiscent of antiquity and the process of B represents the dominance of RNA world.

Choose the option which **correctly** fill in the blanks A and B.

- A – Exons, B – Capping
- A – Introns, B – Splicing
- A – VNTRs, B – Replication
- A – UTRs, B – Translation

162. Select the **correctly** matched pair.

(1)	Exarch type of arrangement of primary xylem	–	Metaxylem lies towards periphery
(2)	Trichomes in plants	–	Usually unicellular and unbranched
(3)	Guard cells	–	Bean-shaped in grasses
(4)	Monocot stem	–	Medullary rays are absent

- (1)
- (2)
- (3)
- (4)

163.In operon model of gene regulation in bacteria, the gene which provides the attachment site for RNA polymerase is

- (1) Structural gene
- (2) Promoter gene
- (3) Regulator gene
- (4) Operator gene

164.Which of the following is/are the salient features of genetic code?

- a. Nearly universal
- b. Overlapping codon
- c. Degeneracy of codon
- d. Commaless nature

- (1) Only b
- (2) Only c and d
- (3) Only a, b and d
- (4) Only a, c and d

165.Cell organelle which is involved in synthesis of lipids and steroids is also involved in

- (1) Proteins synthesis actively
- (2) Processing, packaging and transport of materials
- (3) Providing precursor of enzyme for lysosomes formation
- (4) Detoxification of drugs

166.Select the common features between the cell organelles also called powerhouse of the cell and photosynthetic plastid.

- (a) Double membrane bound
 - (b) ATP synthesis
 - (c) Components of endomembrane system
 - (d) Semi-autonomous organelle
 - (e) Single stranded DNA
- The **correct** ones are

- (1) Only a, b and c
- (2) Only b, d and e
- (3) Only a, b and d
- (4) Only a and d

167.Which of the following sub-stages of cell division represents the transition from prophase I to metaphase I?

- (1) Leptotene
- (2) Diakinesis
- (3) Pachytene
- (4) Zygotene

168.Which of the following events does **not** take place in prophase stage of mitosis?

- (1) Initiation of condensation of chromosomal material
- (2) Each centrosome radiates out microtubules called asters
- (3) Congression of chromosomes at the equator of spindle
- (4) Untangling of DNA strand to form compact mitotic chromosome

169.Match the following columns and select the **correct** option.

Column I	Column II
a. Synthesis of tubulin protein take place	(i) S phase
b. Quiescent stage	(ii) G ₁ phase
c. Interval between mitosis and initiation of DNA replication	(iii) G ₀ phase
d. Duplication of centriole in cytoplasm	(iv) G ₂ phase

- (1) a(i), b(iii), c(iv), d(ii)
- (2) a(iv), b(iii), c(ii), d(i)
- (3) a(iv), b(ii), c(iii), d(i)
- (4) a(ii), b(i), c(iii), d(iv)

170.Which of the following cell organelles is rich in hydrolytic enzymes?

- (1) Golgi bodies
- (2) Lysosomes
- (3) Mitochondria
- (4) Ribosomes

171.An animal having dorso-ventrally flattened body with absence of alimentary canal is

- (1) Filarial worm
- (2) Tapeworm
- (3) *Ascaris*
- (4) Hookworm

172.Which of the following bones is not a part of rib cage in adult humans?

- (1) Thoracic vertebrae
- (2) Sternum
- (3) False ribs
- (4) Cervical vertebrae

173. Globular head of meromyosin has binding sites for _____ and active sites for _____

Choose the **correct** option to fill in the respective blanks.

- (1) Actin, ADP
- (2) Troponin, ATP
- (3) ATP, Actin
- (4) Tropomyosin, Troponin

174. All of the following are present in areolar connective tissue, **except**

- (1) Mast cells
- (2) Fibroblasts
- (3) Chondrocytes
- (4) Adipocytes

175. The type of enzyme that can break the bonds present between monomeric units of insulin, belongs to the class named

- (1) Dehydrogenases
- (2) Lyases
- (3) Ligases
- (4) Hydrolases

176. In which of the following scenarios, MTP **cannot** be performed legally in India?

- (1) To get rid of unwanted pregnancies due to rape
- (2) Failure of the contraceptive used during sexual intercourse
- (3) Continuation of the pregnancy is harmful to the mother
- (4) Foetus is found to be a female

177. Which of the following events will not occur in a tubectomised female?

- (1) Occurrence of menstruation
- (2) Production of estrogen
- (3) Transport of ovum to ampulla
- (4) Formation of ovum

178. Population growth rate increases due to all of the following, **except**

- (1) Decrease in death rate
- (2) Increase in maternal mortality rate
- (3) Increase in number of people in reproductive age
- (4) Increase in medical facilities

179. All of the following are the components of diffusion membrane of human respiratory system, **except**

- (1) Squamous epithelium of alveoli
- (2) Endothelium of alveolar capillaries
- (3) Ciliated columnar epithelium of bronchi
- (4) Basement substance present between epithelium of alveoli and endothelium of alveolar capillaries

180. Select the **incorrect** match w.r.t pulmonary volumes and capacities.

- (1) $TLC = RV + TV + IRV + ERV$
- (2) $FRC = ERV + RV$
- (3) $EC = TV + ERV$
- (4) $VC = ERV + IC + IRV$



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